

1. Administrative & Study Identification

Full Study Title: Cost comparison between apixaban and low molecular weight heparin (LMWH) among venous thromboembolism (VTE) cancer patients **Short Title/Acronym:** Cost Comparison of Apixaban vs LMWH in Cancer-Associated VTE **Protocol Number:** [Not Available / Observational Protocol ID] **NCT Number:** NCT05643885 **Sponsor Name:** Pfizer / Bristol-Myers Squibb (BMS) **Investigational Product:** Apixaban (Eliquis) compared to Low Molecular Weight Heparin (LMWH) **Phase of Development:** Phase 4 / Observational (Non-Interventional) **Medical Monitor:** N/A (Retrospective Claims Database Study)

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To compare the total health care costs between apixaban and low molecular weight heparin (LMWH) in cancer patients with venous thromboembolism (VTE). **Secondary Objectives:** To compare rates of recurrent VTE, major bleeding, and other healthcare resource utilization (HRU) outcomes among VTE cancer patients treated with apixaban versus LMWH.

2.2 Background & Rationale Direct-acting oral anticoagulants (DOACs), such as apixaban, are increasingly recommended and used as an alternative to LMWHs for the treatment of VTE in cancer patients. While clinical trials have shown DOACs to have similar or better efficacy and safety profiles compared to LMWHs, real-world evidence is needed to understand the economic impact and health care costs of using apixaban versus LMWH in this specific high-risk patient population. This study aims to fill that gap by analyzing real-world claims data.

3. Study Design & Methodology

3.1 Design Framework Study Type: Observational (Retrospective Cohort Study) **Control Type:** Active Comparator (LMWH) **Blinding:** Open-label / Non-blinded (Observational) **Randomization:** N/A (Observational study utilizing Inverse Probability of Treatment Weighting [IPTW] to account for baseline confounders)

3.2 Planned Enrollment & Duration Target \$\$\$: [Data dependent on claims database extraction / power calculation] **Number of Sites:** N/A (Retrospective analysis of US healthcare claims databases) **Estimated Study Start:** 2022 **Estimated Primary Completion:** [Estimated 2023/2024]

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Adult patients (≥ 18 years of age). 2. Documented diagnosis of cancer via ICD-10-CM codes. 3. Diagnosis of Venous Thromboembolism (VTE) documented via ICD-10-CM codes. 4. Initiation of apixaban or LMWH during the study identification period. 5. Minimum of 1 month of continuous health plan enrollment post-index date.

4.2 Exclusion Criteria 1. Patients without a confirmed cancer diagnosis prior to or on the VTE index date. 2. Use of multiple anticoagulants simultaneously on the index date. 3. Patients not meeting the required continuous enrollment period criteria.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Standard of care, real-world dosing for Apixaban based on prescription claims. **Route of Administration:** Oral (Apixaban) vs. Subcutaneous Injection (LMWH). **Duration of Treatment:** Variable post-index period (minimum 1 month enrollment).

5.2 Concomitant & Prohibited Therapy Permitted: Standard anti-cancer therapies and concomitant medications for other co-morbidities recorded in the database. **Prohibited:** Initiation of dual anticoagulation on the index date.

6. Study Timeline & Visit Schedule

(Note: As a retrospective claims database study, traditional clinical visits do not apply) | Visit ID | Window (Days) | Key Activities/Procedures | | :--- | :--- | :--- | | **Screening** | Pre-Index | Identification of Cancer and VTE diagnoses via ICD-10-CM codes | | **Baseline** | Day 0 (Index) | First prescription claim of Apixaban or LMWH | | **Interim** | Variable | Tracking of healthcare claims, costs, and clinical events (recurrent VTE/bleeding) | | **End of Study** | Post-Index | End of continuous health plan enrollment or data extraction period |

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Total health care costs (medical and pharmacy costs) during the post-index period, evaluated after adjusting for confounders. **Measurement Method:** Retrospective extraction and analysis of medical and pharmacy claims records.

7.2 Secondary & Exploratory Endpoints 1. Rates of Venous Thromboembolism (VTE) recurrence. 2. Rates of major bleeding and

clinically relevant non-major bleeding events. 3. Healthcare Resource Utilization (HRU) such as hospitalizations and length of stay.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Derived retroactively from ICD-10-CM diagnosis codes (e.g., diagnostic codes for bleeding events). **Serious Adverse Events (SAEs):** Measured as hospitalizations due to major bleeding or VTE recurrence. **Data Monitoring Committee (DMC):** N/A (Observational retrospective study)

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: All eligible patients who initiated apixaban or LMWH with the required continuous enrollment. **Sample Size Justification:** A sample size calculation was conducted to identify the minimum number of patients required to detect a significant difference in healthcare costs. **Handling of Missing Data:** Inverse Probability of Treatment Weighting (IPTW) is utilized to adjust for baseline imbalances and potential confounding between the apixaban and LMWH cohorts.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study is conducted in accordance with Good Pharmacoepidemiology Practices (GPP) and patient data privacy regulations (e.g., HIPAA). **Monitoring Plan:** N/A for retrospective claims analysis. **Audit Trail:** Traceable code algorithms for cohort identification, ICD-10-CM code extraction, and cost calculations.

11. Study Identifiers & Governance

Primary ID: [Internal Sponsor ID / DocuSign ID: A2979DAA...] **Registry Number:** NCT05643885 **Sponsor/Lead Organization:** Pfizer / Bristol Myers Squibb **Study Title:** Cost Comparison between Apixaban and LMWH among VTE Cancer Patients **Official Regulatory Title:** Cost comparison between apixaban and low molecular weight heparin (LMWH) among venous thromboembolism (VTE) cancer patients

12. Clinical Framework (VTE Specific)

Primary Condition: Venous Thromboembolism (VTE) in Cancer Patients **Diagnosis Threshold:** Confirmed ICD-10-CM diagnosis code for VTE (DVT and/or PE) and active cancer **Medical Methodology:** Direct-Acting Oral Anticoagulant (DOAC) vs. LMWH Therapy

Optimization Target: Total healthcare cost reduction and prevention of VTE recurrence

13. Study Procedures & Methodology

Management Pathway: Standard real-world clinical oncology and anticoagulation pathways **Education Protocol (HCP):** N/A (Retrospective Database Study) **Education Protocol (Patient):** N/A (Retrospective Database Study) **Quality Audit Mechanism:** Statistical validation of claims database extraction and IPTW matching

14. Observation & Follow-Up Schedule

Total Duration: Variable post-index tracking (minimum 1 month enrollment) **Onsite Visit Cadence:** N/A **Remote Monitoring Frequency:** N/A (Continuous database extraction) **Checklist Review Interval:** N/A

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				